Learning Optimization

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A learning approach to dynamic pricing
using nonlinear and linear optimization

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The data available at time period t and the approach

- $p_{1,\tau}$: your price levels at time periods $\tau = 1, 2, \ldots, t-1$
- $p_{2,\tau}$: your competitor's price levels at time periods $\tau = 1, 2, \ldots, t-1$
- $d_{1,\tau}$: the amount of demand you sold at time periods $\tau = 1, 2, \ldots, t-1$

1. Estimate your (linear) demand model

$$d_{1,\tau} = \beta^0_{1,\tau} + \beta^1_{1,\tau} p_{1,\tau} + \beta^2_{1,\tau} p_{2,\tau} + \epsilon_{1,\tau}$$

2. Calculate the β-values for an (extrapolated) demand model:

$$d_{1,t} = \beta^0_{1,t} + \beta^1_{1,t} p_{1,t} + \beta^2_{1,t} p_{2,t}$$

3. Estimate your competitor's demand model:

$$d_{2,\tau} = \beta^0_{2,\tau} + \beta^1_{2,\tau} p_{1,\tau} + \beta^2_{2,\tau} p_{2,\tau} + \epsilon_{2,\tau}$$

   assuming your competitor always uses an optimized price
4. Calculate your optimal price level for time period t

Minimize

$$\sum_{\tau=1}^{t-1} \left| d_{1,\tau} - \left(\beta_{1,\tau}^{0} + \beta_{1,\tau}^{1} p_{1,\tau} + \beta_{1,\tau}^{2} p_{2,\tau} \right) \right|$$

with the additional constraints that the β-values do not differ significantly from time period to time period

$$\left| \beta_{1,\tau}^{i} - \beta_{1,\tau+1}^{i} \right| \leq \delta_{1,i}$$

This model can be linearized so that a linear programming solver can be applied:

$$\min \sum_{\tau=1}^{t-1} \Delta_\tau$$

so that

$$\begin{aligned}
d_{1,t} - \left(\beta_{1,\tau}^{0} + \beta_{1,\tau}^{1} p_{1,\tau} + \beta_{1,\tau}^{2} p_{2,\tau} \right) &\leq \Delta_\tau \\
-d_{1,t} + \left(\beta_{1,\tau}^{0} + \beta_{1,\tau}^{1} p_{1,\tau} + \beta_{1,\tau}^{2} p_{2,\tau} \right) &\leq \Delta_\tau \\
\beta_{1,\tau}^{i} - \beta_{1,\tau+1}^{i} &\leq \delta_{1,i} \\
-\beta_{1,\tau}^{i} + \beta_{1,\tau+1}^{i} &\leq \delta_{1,i}
\end{aligned}$$

Step 2: Calculate the β-values for an (extrapolated) demand model

Compute the estimated β-values for the next time period t

Average the β-values for a pre-defined time window N:

$$\beta^i_{1,t} = \frac{1}{N} \sum_{\tau=t-N}^{t-1} \beta^i_{1,t}$$

With these β-values you can estimated your next sales given your price level and the estimated competitor's price level.

BHT

Assume that your competitor uses an optimal price

The optimal price level would be

$$p_{2,t} = \frac{\beta^0_{2,t} + \beta^1_{2,t} p_{1,t}}{-2\beta^2_{2,t}}$$

where the β-values are unknown but can be estimated by minimizing

$$\sum_{\tau=1}^{t-1} \left| p_{2,\tau} - \left(\frac{\beta^0_{2,\tau} + \beta^1_{2,\tau} p_{1,\tau}}{-2\beta^2_{2,\tau}} \right) \right|$$

once again assuming that the β-values do not differ significantly from time period to time period

$$\left| \beta^i_{2,\tau} - \beta^i_{2,\tau+1} \right| \le \delta_{2,i}$$

BHT

Compute the estimated β-values for the next time period t

Average the β-values for a pre-defined time window N:

$$\beta^i_{2,t} = \frac{1}{N} \sum_{\tau=t-N}^{t-1} \beta^i_{2,t}$$

With these β-values you can estimated the next sales of your competitor given your price level and the estimated competitor's price level.

Step 4: Calculate your optimal price level for time period t

Your optimum price

Assuming that your competitor will use an optimized price, i.e.

$$p_{2,t} = \frac{\beta^0_{2,t} + \beta^1_{2,t} p_{1,t}}{-2\beta^2_{2,t}}$$

your optimal price should maximize

$$p_{1,t} d_{1,t} = p_{1,t} \left(\beta^0_{1,t} + \beta^1_{1,t} p_{1,t} + \beta^2_{1,t} p_{2,t} \right)$$

This is an one-dimensional function in $p_{1,t}$ that can be maximized using standard solvers:

$$\max p_{1,t} \left(\beta^0_{1,t} + \beta^1_{1,t} p_{1,t} + \beta^2_{1,t} \frac{\beta^0_{2,t} + \beta^1_{2,t} p_{1,t}}{-2\beta^2_{2,t}} \right)$$